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SARNet : Speaker-Attentive ResNet for Quantification of Dysarthria Severity

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Abstract:

Analyzing differences in audio data of individuals with articulation disorders from the perspective of human speech and employing objective methods for automated dysarthria evaluation can significantly aid doctors in early patient screening and diagnosis. This proactive approach enables timely intervention and treatment during the initial stages of the condition. This paper presents a speech recognition network(named Speaker-Attentive ResNet (SARNet)) that aggregates and propagates features from different hierarchical levels using an attention-based statistical pooling module, built upon the ResNet architecture. This network aims to extract subtle features specific to individual speakers. The proposed method is evaluated using the TORGO dysarthric speech database, employing eight different acoustic features as well as features aggregated from these eight. Comparative experiments with two other models, Time-Delay Neural Network (TDNN) and Panns-CNN10(Large-Scale Pretrained Audio Neural Networks), demonstrate that due to the superior ability of MFCC(Mel-scaleFrequency Cepstral Coefficients) to emulate human auditory features, the proposed approach achieves a classification accuracy of 98%-99% in measuring speech intelligibility in healthy people, patients, and the severity of patient's articulation disorders.

Key Words: Dysarthria, Quantification, Speaker-Attentive ResNet

1 Introduction

Dysarthria refers to a speech disorder caused by organic lesions of the neuromuscular system, resulting in weak or paralyzed muscles of the vocal organs, or abnormal muscle tone and uncoordinated movement, leading to abnormal voice production, articulation, resonance, rhythm, and unclear speech[1].

Dysarthria is common in conditions such as cerebrovascular accidents, cerebral palsy, amyotrophic lateral sclerosis, Parkinson's disease,and others[2]. Dysarthria can be classified based on the presentation of symptoms, including hypokinetic, hyperkinetic, ataxic, spastic-athetoid, spastic, and flaccid types[3, 4].

Intelligibility of speech has been used as a general indicator of the severity of speech disorders[5], including dysarthria[6]. Intelligibility and speaking rate can be used as quantitative measures of the severity of dysarthria[7]. The assessment of intelligibility in patients with speech disorders is performed by speech-language pathologists through auditory perception tests, which rely on the subjective judgment and invasive diagnosis of the physician[8–11].

In the field of acoustics, speech processing used many perceptual features, such as linear prediction coefficients, Mel-frequency cepstral coefficients (MFCCs), constant Q cepstral coefficients(CQCCs) and perceptual linear prediction coefficients. Among them, MFCCs have demonstrated their usefulness in modeling pathological speech signals[12] in general and in identifying dysarthria severity using machine learning classifiers in specific cases[13, 14]. In [14], specific prosodic features such as spectral moments, formants, skewness and MFCCs were selected by genetic algorithm and used together with support vector machine(SVM) classifiers for speech diagnosis and severity recognition of dysarthria. Chandrashekar et al. aimed to evaluate the intelligibility of speech produced by dysarthric patients using different time-frequency representation methods(like perception-enhanced Fourier transform spectrograms and constant Q-transform spectrograms)[15]. G. Tartarisco et al. proposed a novel hierarchical machine learning model(HMLM) to improve the detection and assessment of dysarthria in structured speech interference tests for children with ataxia[16]. In terms of model, Kim et al. proposed a structured sparse linear model integrating speech knowledge, which simultaneously addressed phonologically structured sparse feature selection and intelligibility prediction[17]. In[18], a machine-learning framework with a robust three-step feature selection criterion and a Bayesian data-driven clustering technique based on the multivariate mixture extension of the generalized linear mixed model(GLMM) was adopted to objectively predict the progression of Cerebellar dysarthria (CD) in cerebellar ataxia (CA) patients over time. Joshy et al. explored the utility of low-dimensional feature representation using subspace modeling to give i-vectors and grading the severity of dysarthria using a DNN model[19].

Drawing inspiration from successful implementations, this paper focuses on aiding doctors in early screening and diagnosis of patients by using an objective method to automatically evaluate dysarthria, quantify patient severity, and enhance evaluation accuracy. This approach facilitates early intervention and treatment for patients, as well as enables the assessment of treatment efficacy over the course of therapy.

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The paper presents a speech recognition network[20] named Speaker-Attentive ResNet (SARNet), which aggregates and propagates features from different hierarchical levels using an attention-based statistical pooling module, built upon the ResNet architecture. This network is designed to extract subtle features specific to individual speakers. Furthermore, the paper conducts a comparative analysis of eight traditional acoustic features and concatenated aggregate features to determine the optimal classification combination, thereby enhancing the accuracy of speech intelligibility evaluation.

The rest of the paper is organized as follows. Section 2 introduces the TORGO database, acoustic feature extraction, deep learning methods and evaluation metrices used in this work. Section 3 presents the experimental results and comparative analysis of this paper. Finally, Section 4 provides a summary of the entire article.

2 Methods

2.1 Database

The TORGO database[21] of dysarthric articulation consists of aligned acoustics and measured 3D articulatory features from speakers with either cerebral palsy(CP) or amyotrophic lateral sclerosis(ALS). TORGO database consists of 8 dysarthric(DYS) speakers(3 females and 5 males) and 7 non-dysarthric or healthy control(HC) speakers(3 females and 4 males) as a control group. The age of the patients ranged from 16 to 50. Speaker IDs beginning with F represent female speakers and ones that begin with M represent male speakers. The acoustic data were recorded through two different microphones; a microphone array with 8 recording elements placed at a distance of 61 cm facing the speaker, and a head-mounted microphone. The motor functions of each subject were assessed according to the standardized Frenchay Dysarthria Assessment (FDA)[22] by a speech-language pathologist.

The severity levels are assigned as very low, low, medium and high, based on clinical intelligibility and articulatory functionality by an SLP , as per FDA. The intelligibility rating of each severity level is as follows:(0-25)%-high, (25-50)%-medium, (50-75)%-low and (75-100)%-very low. The population was divided into five categories according to the portion of Part about speech in the MDS-UPDRS[23] rating scale, with higher scores representing worse illness. Description of the databases is given in Table 1.

Table 1: Subject Wicse Intelligibility Distribution and Grading Situation for TORGO Database

Severity	Speech Intelligibility	TORGO	Score
VERY LOW	(75-100)%	F03,F04,M03	1
LOW	(50-75)%	F01,M05	2
MEDIUM	(25-50)%	M01,M02,M04	3
HIGH	(0-25)%	-	4
HEALTHY	100%	FC01, FC02, FC03 MC01, MC02 MC03, MC04	0

This paper uses all Short words audio data collected with a head-mounted microphone with a total of 5869 wav files ranging from 1 to 5 seconds, with control data volume of 3716 and patient data volume of 2153, as shown in Table 2. While using this database, 80% of the data is used for training and the rest for testing.

Table 2: Details of TORGO Dysarthric Speech Database

Severity	TORGO(Short words)	Sum
VERY LOW	F03:413 F04:180 M03:309	Total:2153
LOW	F01:94 M05:373	
MEDIUM	M01:279 M02:297 M04:208	
HIGH	-	
HEALTHY	FC:1537 MC:2179	Total:3716
Total		5869

2.2 Acoustic Feature Extraction

This article explores the impact of SpectralCentroid[24], SpectralBandwidth, SpectralRollOff, ChromaStft, Spectrogram, MelSpectrogram, LFCC, MFCC, and their aggregate features FeatureAggregation on evaluation accuracy. Taking Spectrogram and MFCC as examples of acoustic features, they are briefly introduced. The features are extracted using functions in the Librosa library and used as inputs for subsequent network models. In the process of processing audio signals, operations such as pre-emphasis, framing, windowing, and fast Fourier transform will be performed, as shown in Fig. 1.

- **Spectrogram**. It is necessary to convert signals into corresponding spectrograms, which represent the features of the signal on a two-dimensional plane. The energy values are represented using colors, with darker colors indicating higher energy levels of the speech signal.
- **Mel-Frequency Cepstral Coefficients(MFCCs)**. Due to the natural sensitivity of the human auditory system to low-frequency information and the relative insensitivity to high-frequency information, the Mel-frequency cepstral coefficients(MFCC) are designed to mimic the human ear's auditory system. They constitute the mel-frequency cepstrum(MFC), a compact representation of the short-term power spectrum of an audio signal, obtained through a linear cosine transform from the log power spectrum to the nonlinear mel scale frequency[25].

In the training process, we need to perform data augmentation operations on the audio data in the TORGO Database. Following the normalization requirements, after reading the audio data, we crop the silent segments and keep the non-silent segments in the audio signal. The audio length is uniformly set to 6 seconds, and the audio shorter than 6 seconds is padded, while the audio longer than 6 seconds is truncated. All audio data is downsampled to 16000Hz, and the target decibel level is set to normalize the audio volume, so as to enhance the training data and improve the accuracy of the model training.

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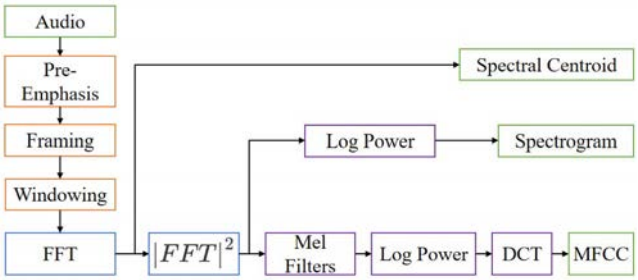


Fig. 1: the General Steps of Audio Feature Extraction

2.3 Classification Model

We leverage the advantages of ECAPA-TDNN[20]. We construct the proposed network structure, Speaker-Attentive ResNet(SARNet), on the basis of ResNet network. Aggregating and propagating features from different levels in the encoder to capture shallow and deep speaker feature maps; Enhancing the statistical pooling module based on channel and context-aware attention frameworks to focus more on individual speech characteristics, such as paying attention to more vowels rather than consonants.

According to the network's multi-layer feature aggregation and channel/context-aware statistical pooling, after passing through SARNet, the input audio is transformed into an embedding vector. This embedding vector effectively represents the features of input audio and possesses discriminative capability. Fig. 2 illustrates the framework of our method for quantifying the severity of dysarthria in patients based on the SARNet network.

2.4 Evaluation Metrics

In paper, we utilize accuracy, precision, recall, and F1 score to evaluate performance of the proposed modelc(SARNet). Below are the corresponding evaluation metrics and their formulas:

- **Accuracy**
  The accuracy is defined as the ratio of correctly classified instances to the total number of instances:
  $$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{1}$$
  where TP represents true positives, TN true negatives, FP false positives, and FN false negatives.
- **Precision**
  Precision measures the proportion of positive identifications that were actually correct:
  $$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{2}$$
- **Recall (Sensitivity)**
  Recall, also known as sensitivity, measures the proportion of actual positives that were identified correctly:
  $$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{3}$$
- **F1 Score**
  The F1 score is the harmonic mean of precision and recall:
  $$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$

3 Results

This section describes the relevant parameter settings of experiments. Acoustic processing software and related libraries are utilized to extract features from audio recordings of healthy individuals and patients(using Spectrogram and MFCC as examples), and the features are visualized to characterize them. The focus of the experiment is on using three different deep learning techniques(TDNN, PANNS-CNN10, and SARNet) to compare and analyze the accuracy of quantifying severity of patient pathology using eight traditional acoustic features and their aggregated feature(Feature Aggregation). This provides support for automatic speech recognition technologies for pathological speech.

3.1 Experimental Setup

- The standard American English dysarthric database, namely the TORGO[21] database is used for evaluating the proposed work. All speech samples were downsampled to 16kHz and used to extract relevant features, which were either 513-dimensional spectrograms or 40-dimensional MFCCs or 1-dimensional Spectral Centroid(Spectral Bandwidth and Spectral Rolloff are both one-dimensional feature dimensions.)or 20-dimensional LFCC or 12-dimensional ChromaStft or 64-dimensional Melspectrogram or 632-dimensional FeatureAggregation, as described in paper.
- We implemented TDNN, PANNS-CNN10, and SARNet as classification networks for training. After extraction, the final embedding dimensions for the three models were 192, 512, and 192, respectively.
- All models were optimized using the Adam optimizer[26] with the Cosine Annealing learning rate decay method, gradually reducing the learning rate during the training process to improve model training performance. The duration of one cycle is set to 100 iterations. The batch size for training is 32.
- All models use AttentiveStatsPool as the pooling layer, which extracts statistical features from the input sequence using attention mechanism and obtains the mean and standard deviation of the input sequence through a series of operations. Then other operations are performed based on these statistics.

3.2 Features Visualization

As we all know, there are significant differences between the speech of healthy individuals and those with articulation disorders. To objectively represent these differences beyond human perception, let's consider three female speakers who differ in their speech ratings but read the same word. Using the librosa[27] library and Praat[28] software, we will extract two acoustic features: Spectrogram and MFCC, to characterize these features.

Select audio data from three female speakers, F01, F03, and FC01, for analysis. FC01 is chosen as the healthy control group. All three speakers read the word "pat". Use the Praat[28] software to plot the Spectrogram feature graph, where the horizontal axis represents time, the vertical axis represents frequency, and the coordinate point value represents the energy of speech data. The magnitude of the energy value is represented by the color, with darker colors indicat-

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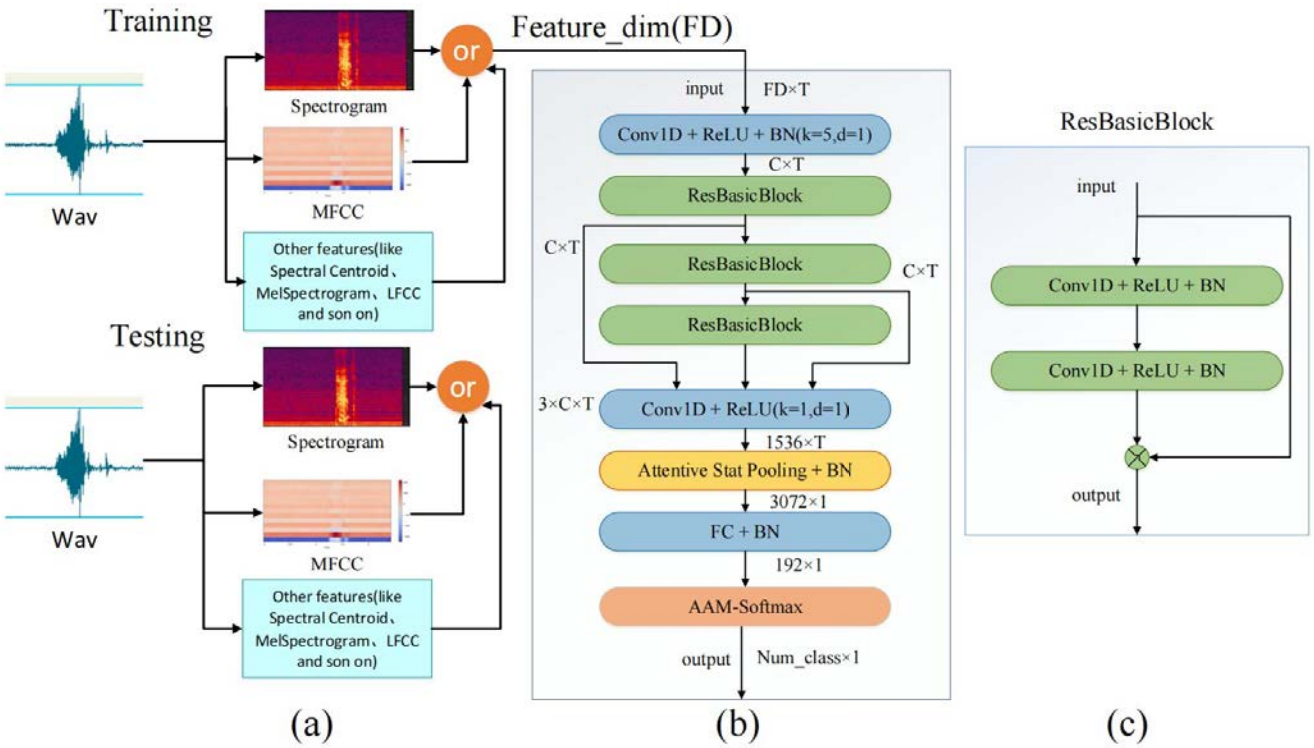
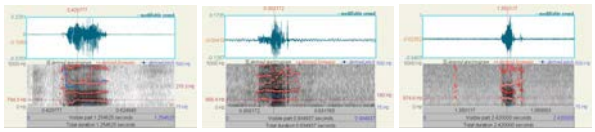


Fig. 2: A framework for quantifying the severity of dysarthria in patients has been developed based on the SARNet. (a) depicts the feature extraction process during training and testing. (b) shows the whole network of SARNet. K stands for kernel size and d for dilation. C and T denote channels and temporal dimensions. (c) is the detailed illustration of ResBasicBlock in the SARNet model.

ing stronger speech energy at that point. Spectrograms compensate for the loss of temporal information in frequency spectrum, so they are presented in a two-dimensional form, encompassing three-dimensional information: (vertical axis: frequency; horizontal axis: time; color depth: amplitude value magnitude).

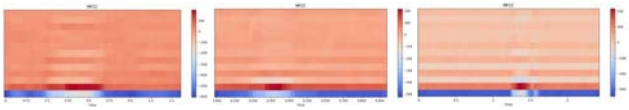


(a) F01(Score:2) (b) F03(Score:1) (c) FC01(Score:0)

Fig. 3: Visualization of Spectrogram feature

In Fig. 3, for the dysarthria samples it can be seen that the energy magnitudes of the frequencies are more spread out across time, indicative of slow and slurred speech or due to the fact that the words are coming out more rapidly and are overlapping with each other. For the Non dysarthria samples it can be seen that the energy magnitudes are more narrowed to only the parts where they have spoken something, which is regular paced.

Next, draw MFCC feature maps for three subjects to observe differences between them. The main idea behind MFCC is to simulate perceptual characteristics of the human ear to sound. The human auditory system is sensitive to low-frequency information and relatively insensitive to high-frequency information. MFCC has good simulation effects on the frequency and time domain characteristics of human speech perception.



(a) F01(Score:2) (b) F03(Score:1) (c) FC01(Score:0)

Fig. 4: Visualization of MFCC feature

From Fig. 4, we observe that for dysarthria samples, the MFCC visualization has much more darker and concentrated pixels indicating the slurs present in their speech. For non dysarthria it is not as concentrated and is more spread out.

3.3 Comparison

After all preparations are completed, this paper uses precision, recall, f1-score, and accuracy to evaluate the experimental results. Nine acoustic features, including spectrograms and MFCC, are used in the three deep learning networks mentioned above. The goal is to find the best combination of acoustic features and deep learning techniques, greatly improving the quantification accuracy and providing more accurate diagnostic results for doctors.

The results will be analyzed from two aspects. Firstly, we need to distinguish between healthy individuals and patients with dysarthria, and ensure that this result is as close to reality as possible. Secondly, regarding the intelligibility evaluation of speech from patients with dysarthria, we need to improve the quantification accuracy based on existing technology. Table 3 and Table 4 show the classification results of the healthy control group and patients with varying degrees of illness, respectively.

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Table 3: Health classification

Network		Tdnn	Panns-CNN10	SARNet
Spectral Centroid	precision	0.883	0.885	**0.892**
	recall	0.883	0.882	**0.892**
	f1-score	0.882	0.879	**0.891**
	accuracy	0.883	0.882	**0.892**
Spectral Bandwidth	precision	0.890	0.898	**0.905**
	recall	0.888	0.895	**0.903**
	f1-score	0.885	0.893	**0.901**
	accuracy	0.888	0.895	**0.903**
Spectral RollOff	precision	0.887	0.883	**0.902**
	recall	0.882	0.880	**0.902**
	f1-score	0.879	0.877	**0.902**
	accuracy	0.882	0.880	**0.902**
Chroma Stft	precision	0.925	**0.941**	0.938
	recall	0.923	**0.940**	0.938
	f1-score	0.922	**0.940**	0.938
	accuracy	0.923	**0.940**	0.938
Spectrogram	precision	0.964	**0.979**	0.966
	recall	0.964	**0.979**	0.966
	f1-score	0.964	**0.979**	0.966
	accuracy	0.964	**0.979**	0.966
Mel-Spectrogram	precision	0.966	0.950	**0.972**
	recall	0.966	0.950	**0.972**
	f1-score	0.966	0.949	**0.972**
	accuracy	0.966	0.950	**0.972**
LFCC	precision	0.986	0.976	**0.989**
	recall	0.986	0.976	**0.989**
	f1-score	0.986	0.976	**0.989**
	accuracy	0.986	0.976	**0.989**
MFCC	precision	**0.991**	0.986	0.985
	recall	**0.991**	0.986	0.985
	f1-score	**0.991**	0.986	0.985
	accuracy	**0.991**	0.986	0.985
Feature Aggregation	precision	**0.989**	0.952	0.988
	recall	**0.989**	0.951	0.988
	f1-score	**0.989**	0.951	0.988
	accuracy	**0.989**	0.951	0.988

In Tables 3 and 4, bold text in each row indicates the best-performing model among the three neural network models under the same acoustic feature input. The gray background numbers in each column represent the best-performing acoustic feature obtained from comparing nine different acoustic features under the same deep learning model.

From Table 3, it can be seen that the precision, recall, f1-score, and accuracy of SpectralCentroid, SpectralBandwidth, and SpectralRollOff are all around 90%. The reason for this is that these three acoustic features extract sound information from a frequency perspective and do not comprehensively consider other information such as time domain. Therefore, using them to achieve classification accuracy is not very high.

The remaining six features extract speech features from both frequency and time domains. In terms of classification accuracy, the LFCC and MFCC features perform better, achieving 98%-99% accuracy across the three different network models. As for FeatureAggregation, although it also achieves good accuracy in the three models, it is still slightly lower. We speculate that the reason may be that the selected features have a strong correlation with each other, causing the aggregated features to lose some important information, resulting in inaccurate classification results.

Table 4: Classification of Disease Severity

Network		Tdnn	Panns-CNN10	SARNet
Spectral Centroid	precision	0.827	0.813	**0.863**
	recall	0.832	0.821	**0.867**
	f1-score	0.829	0.810	**0.864**
	accuracy	0.832	0.821	**0.867**
Spectral Bandwidth	precision	0.821	0.807	**0.861**
	recall	0.826	0.818	**0.865**
	f1-score	0.820	0.807	**0.861**
	accuracy	0.826	0.818	**0.865**
Spectral RollOff	precision	0.832	0.822	**0.878**
	recall	0.837	0.830	**0.882**
	f1-score	0.832	0.818	**0.879**
	accuracy	0.837	0.830	**0.882**
Chroma Stft	precision	0.906	0.891	**0.920**
	recall	0.908	0.892	**0.921**
	f1-score	0.906	0.891	**0.919**
	accuracy	0.908	0.892	**0.921**
Spectrogram	precision	0.951	**0.971**	0.961
	recall	0.951	**0.971**	0.962
	f1-score	0.951	**0.971**	0.961
	accuracy	0.951	**0.971**	0.962
Mel-Spectrogram	precision	0.950	0.902	**0.957**
	recall	0.950	0.902	**0.957**
	f1-score	0.949	0.901	**0.957**
	accuracy	0.950	0.902	**0.957**
LFCC	precision	0.956	0.958	**0.969**
	recall	0.956	0.958	**0.968**
	f1-score	0.955	0.958	**0.968**
	accuracy	0.956	0.958	**0.968**
MFCC	precision	0.979	0.975	**0.983**
	recall	0.979	0.975	**0.983**
	f1-score	0.979	0.975	**0.983**
	accuracy	0.979	0.975	**0.983**
Feature Aggregation	precision	0.980	0.914	**0.981**
	recall	0.980	0.913	**0.981**
	f1-score	0.980	0.913	**0.981**
	accuracy	0.980	0.913	**0.981**

In Table 4, overall, the accuracy of disease severity classification is lower than that of healthy people and patient classification accuracy. Among them, using MFCC features in the SARNet model, the accuracy is 98.3%, while using MFCC in the Panns-CNN10 model, the accuracy is 97.5%. The accuracy of FeatureAggregation in Tdnn is 98.0%. This indicates that combining MFCC with the proposed model can achieve better evaluation accuracy, greatly improving the diagnostic accuracy of patients and helping doctors implement more effective treatment measures.

4 Conclusion

This paper constructs a speech recognition network, SARNet, capable of extracting subtle features specific to individual speakers. It compares and analyzes three different

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deep learning models, primarily SARNet, with commonly used acoustic features in audio signal processing: Spectrogram, MelSpectrogram, SpectralCentroid, SpectralBandwidth, Spectral Rolloff, ChromaStft, LFCC, MFCC, and their aggregated features(Feature Aggregation). The paper distinguishes healthy people from patients in the TORGO DATABASE, while also evaluating the speech intelligibility of patients with articulatory disorders. Through comparative analysis, it was found that features only related to the spectrum (such as SpectralCentroid) reflect very few sound characteristics and do not help much in quantifying the degree of illness. On the other hand, other features such as MFCC, Spectrogram, and Feature Aggregation, which reflect the multi-level features of sound, can effectively evaluate speech intelligibility. It has been observed that the integration of the proposed network with MFCC features resulted in a significant enhancement in classification accuracy. It is anticipated that this method can assist doctors in early patient screening and diagnosis, enabling timely intervention in the early stages of the disease progression.

Limitations. The research in paper is based on the English language corpus TORGO database. However, this study does not include data from Chinese-speaking patients with articulation disorders. In the future, we will focus on collecting Chinese pronunciation data to enhance the robustness of the proposed model.

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